
Working Memory in the Chain of Thought Is Not Captured by J-Space

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Abstract

1 A language model’s internal workspace supports at most a few steps of chained
2 reasoning, at every scale we test, and the chain of thought holds the rest. A com-
3 mon reading of J-space, the concept workspace a fitted Jacobian lens reads from
4 the residual stream, is that reasoning is carried internally and writing relieves the
5 workspace by externalizing state it would otherwise hold. Three measurements
6 contradict the relief picture. Instructed to answer chained arithmetic without writ-
7 ing, models from 1.5B to 671B parameters collapse beyond one to two dependent
8 steps; a 671B model, so instructed, fails two chained additions. Models write ev-
9 ery intermediate value from the first step at every difficulty, so writing shows no
10 onset threshold. The allocation between the two sides never moves: damaging
11 the workspace does not increase writing, capping writing does not move compu-
12 tation inward, and ablating the model’s active J-space concepts at the strongest
13 dose that preserves ordinary generation changes nothing, even at the one-to-two-
14 step depths where the workspace does perform the arithmetic: the memory the
15 computation uses is not captured by that readout. We then show how this memory
16 works, with causal interventions on six unmodified models (used as released, with
17 no fine-tuning or task-specific training) from four families on arithmetic and nar-
18 rative state-tracking tasks. When a model writes an intermediate value, the hidden
19 state under that number in the text carries the value: overwriting it with the hidden
20 state a different number would have produced, text unchanged, redirects the final
21 answer to the different number in 0.70 to 1.00 of cases, while size-matched ran-
22 dom noise redirects at most 0.01. Later steps retrieve the value by attending back
23 to that number (blocking only that attention drops the answer’s dependence on
24 it to at most 0.007, with matched controls unchanged), the retrieval completes by
25 two thirds of the network’s depth, and two written values can be replaced indepen-
26 dently, the planted pair composing into a sum that appears nowhere in the input
27 (0.88 to 0.90). For oversight, identical visible text can be made to carry different
28 stored values with different downstream behavior: a chain of thought exposes the
29 symbols of a computation, without guaranteeing their hidden content.

30 1 Introduction

31 The chain of thought is the primary working memory of serial reasoning in language models; the
32 internal workspace behind it supports at most a few dependent steps, and writing is not an inference-
33 time overflow from it. We establish this claim with causal interventions on six unmodified models
34 from four families, on two task families with exact ground truth for every intermediate value.

35 The capacity half addresses the overflow picture of externalization: reasoning state is carried inter-
36 nally, and writing takes over when the workspace can no longer hold it. This was our pre-registered

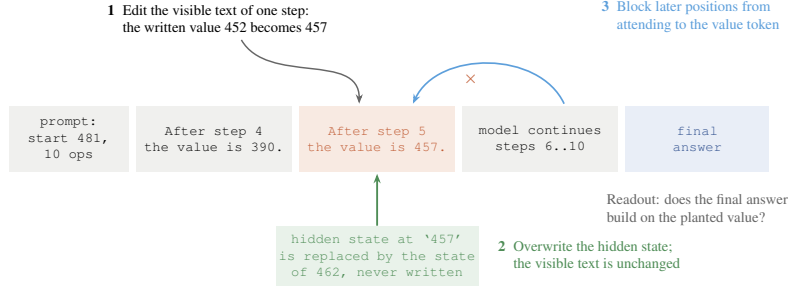


Figure 1: The edit-and-continue protocol. One written step of a correct worked solution is altered and the model continues from it; the readout is whether the final answer builds on the altered value. The interventions: (1) edit the visible text of the step; (2) overwrite the hidden state at the value’s token, leaving the text unchanged; (3) block attention from later positions to the value’s token.

hypothesis, and it is a natural reading of workspace accounts of reasoning, including J-space, the concept workspace a fitted Jacobian lens reads from the residual stream [Gurnee et al., 2026]. A workspace that writing relieves should show up in three places; it shows up in none.

1. **No fallback.** Instructed to answer chained arithmetic with no writing (token counts confirm the channel is closed), models collapse beyond one to two dependent steps at every scale to 671B (Section 3).
2. **No threshold.** Models write 0.93 to 1.00 of ground-truth intermediates from one-step problems onward, at every scale from 1.5B to 671B (Section 3).
3. **No reallocation.** Damaging the workspace does not increase writing, capping writing does not move computation inward, and ablating the top active J-space concepts at a generation-preserving dose changes neither accuracy nor what gets written (Section 3).

The second half is the mechanism of the memory that remains.

4. **Storage.** The hidden state under a written number carries the value the model later uses: planting a never-written value there, text unchanged, redirects the answer at 0.70 to 1.00 across all six models (Section 4).
5. **Retrieval.** Later steps retrieve the value by attending back to the written number: blocking only that attention makes the answer independent of the value (at most 0.007 on four models in three families, with matched control blocks unchanged); layer sweeps time the retrieval to the middle third of the network (Section 5).
6. **Composition, and when models trust what they wrote.** Two positions are independently addressable, composing planted values into their sum (0.88 to 0.90). Which prompt evidence triggers recomputation is model-specific, and reliance on the trace tracks whether the model can regenerate the state, reconciling conflicting reports on how trace reliance scales (Sections 4 and 6).

Prior work anticipated the memory role of written tokens theoretically [Merrill and Sabharwal, 2024], behaviorally [Lanchantin et al., 2023], and causally on a fine-tuned model [Shih et al., 2026] and on multiplication and dynamic-programming tasks [Zhu et al., 2025], without testing whether an internal alternative exists, how the value is retrieved, or whether stores compose.

2 Experimental setup

2.1 Tasks and models

Both task families give exact ground truth for every intermediate value, so the downstream effect of any edit is computable. Arithmetic chains start from a three-digit number and apply d signed two-digit additions and subtractions. Narrative state-tracking problems describe a character whose count

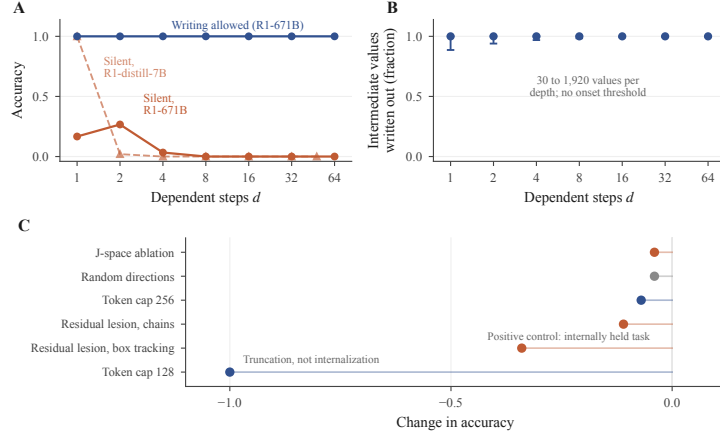


Figure 2: Writing is not triggered by exhausted internal capacity. (A) Instructed to answer without writing, models collapse after one to two dependent steps; with writing allowed, accuracy stays near ceiling ($n=30$ items per depth). (B) The fraction of intermediate values written is 1.00 at every depth (Wilson intervals); no onset threshold appears. (C) No intervention moves state between the workspace and the trace: J-space ablation and random directions leave accuracy unchanged; residual lesions hurt the internally held box-tracking task three times more than the written chain task; a token cap below the per-step floor collapses accuracy by truncation.

of an item changes through varied events and surface forms. In both, the model writes one line per step giving the running value. White-box experiments use Qwen3-4B, Qwen2.5-7B-Instruct, Phi-3-medium, OLMo-2-7B-Instruct, and DeepSeek-R1-Distill-Qwen 7B and 14B; frontier models (DeepSeek V3.2, Claude Sonnet 4.5, Llama 3.3 70B) are tested behaviorally through assistant prefill.

2.2 Edit-and-continue protocol

The protocol takes a correct worked solution, changes the value at one step j , truncates immediately after the edited line, and lets the model continue (Figure 1). A no-edit truncation measures the sampling noise floor, 0.00 throughout. Generation uses temperature 0.6 and top- p 0.95 with five samples per item; unparseable outputs are counted separately. Intervals are 95 percent bootstrap intervals over items. Prompts and configurations are in the appendix; code and data will be released.

3 Does writing replace internal working memory?

3.1 Performance without written intermediates

Instructed to output only the final answer (Figure 2A), with token counts confirming nothing else was generated, models fail beyond one to two dependent steps (Qwen2.5-7B: 1.00 at $d = 1$, 0.04 at $d = 4$; DeepSeek V3.2 at 671B: 0.23 at $d = 1$, near zero above). The values are random, so nothing can be recalled; the failures bound performance under this instruction, with its prompting and training-distribution effects, rather than latent internal capacity.

3.2 When models write intermediate values

Models write 0.93 to 1.00 of ground-truth intermediates in free generation from $d = 1$ through $d = 64$, at every scale from 1.5B to 671B: no difficulty threshold at which writing begins is observed. This rules out an inference-time switch from internal holding to writing; an always-write policy learned in training could itself descend from capacity limits.

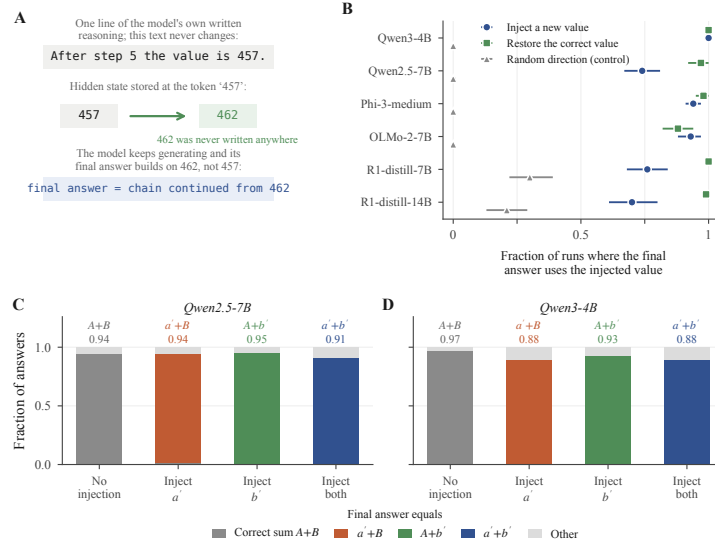


Figure 3: The hidden state under a written number carries the value the model later uses, and two planted values compose. (A) The text still says 457; the hidden state at that token now encodes 462, which appears nowhere, and the final answer continues from 462. (B) Fraction of runs whose final answer uses the injected value; bars are 95 percent bootstrap intervals. The reasoning-distilled models re-solve after their reasoning phase, raising their random-control rate; re-derivation can only produce the correct answer. (C, D) With two counters A and B , injecting a replacement for either or both (a' , b') makes the answer equal the matching sum; answers using only one of two injected values occur at 0.01.

3.3 Tests for reallocation between internal and written state

A residual-stream lesion strong enough to cross the accuracy cliff makes traces disintegrate rather than expand; hard token budgets compress wording up to 2.5 times while keeping 0.98 to 1.00 of values, then truncate below roughly twelve tokens per step; and ablating J-space [Gurnee et al., 2026] changes nothing at the strongest dose preserving ordinary generation (next-token agreement 0.87, perplexity ratio 1.12). The J-space ablation removes the top 16 active concepts at every position and leaves chain-of-thought accuracy, bare-answer accuracy, and the amount written unchanged, on synthetic chains and on GSM8K, doing no more than a size-matched random-directions ablation. The null holds at one to two steps, where models do perform the arithmetic internally.

This bounds the particular readout tested rather than the workspace as a whole: the dose is capped at generation-preserving strength, and a written token's contextual representation and its J-space component are not exclusive alternatives, since the lens samples the same residual states our patches overwrite. The workspace also exceeds J-space by construction: attention and MLP interiors and the KV cache fall outside the lens's domain, and within the residual stream the lens spans a fitted concept dictionary rather than the full state. Section 4 provides an existence proof of the gap: a value planted in the hidden state under a written number is causally decisive, sits in residual states the lens samples, and survives removal of the top 16 concepts.

4 Written tokens store causally active intermediate values

4.1 Restoring and replacing stored values

Figure 3 decomposes the effect of editing a written value. Restoring the correct hidden state while the text still shows a corrupted value returns the answer to correct in 0.88 to 1.00 of cases. Planting a third value that appears nowhere in the text redirects the answer to it in 0.70 to 1.00 of cases. The size-matched random control sits at or below 0.01 for the four instruction-tuned models. The patch replaces the position's full contextual state rather than an isolated numeric component; because answers track whichever value's state is planted, the value that state carries is the causal quantity,

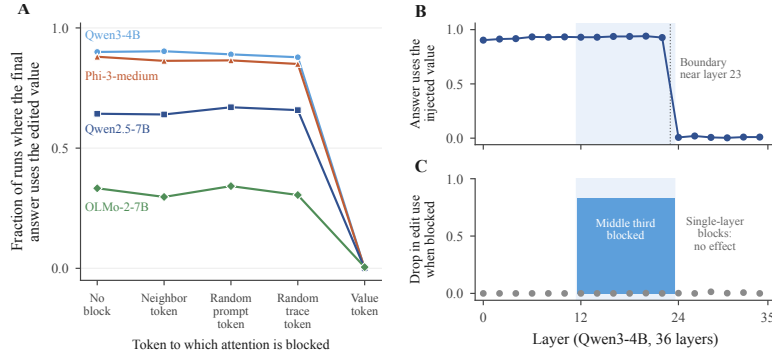


Figure 4: Written values are retrieved through attention to their token, across the middle third of layers. (A) Blocking later positions’ attention to the edited value’s token drops edit use to zero on four models; matched control blocks change nothing ($n=600$ per condition). (B) Overwriting the hidden state at any single layer up to 22 redirects the answer; from layer 24 on it does nothing (Qwen3-4B, 36 layers, $n=60$). (C) Blocking attention across layers 12 to 23 (shaded in both panels) removes 0.83 of edit use; no single layer’s block matters (gray dots, $n=80$).

and because the planted value is a mid-chain intermediate the model never produced, the patch does not inject the answer. The effect persists to 40-step chains (0.98 to 0.95 on Qwen3-4B). Rates are conditioned on items where the text edit was followed (0.44 to 0.98 by model; full table in the appendix).

On the narrative task the pattern replicates: a planted never-written count is followed at 0.83 [0.75, 0.90] on Qwen3-4B and 0.86 [0.79, 0.92] on Phi-3-medium, restoring the correct state returns the answer at 0.89 and 0.94, and the random control is 0.00 in both ($n = 86$ and 89).

4.2 Independent storage and composition

In a task with two counters summed at the end (Figure 3C, D), planting a never-written value at one final position yields the corresponding mixed sum (0.94 and 0.93 on Qwen2.5-7B, 0.89 and 0.93 on Qwen3-4B). Planting both yields their joint sum at 0.90 and 0.88, at or above the product of the single-position rates (0.89 and 0.83). The random patch leaves the clean answer in place (0.96 and 0.95), and answers using only one of the two planted values occur at 0.01. The reported number must be assembled at answer time from two values set independently and never written anywhere.

5 How models retrieve written state

5.1 Attention-blocking experiments

Blocking attention from all post-edit positions to the edited value’s position collapses the answer’s dependence on that value on every model tested (Figure 4A); matched blocks of the adjacent words, a random prompt position, or a random trace position leave it at the unblocked level. Blocking the operand the next step needs breaks the arithmetic itself. The collapse holds across three families and both task formats (natural-language: 0.730 to 0.007 and 0.720 to 0.003 on Qwen3-4B and Phi-3-medium).

Under the value-position block the unparseable-output rate stays at 0.000 (OLMo-2-7B: 0.040, its unblocked rate). The model produces a fluent, parseable number matching neither value: it computes forward from state it can no longer retrieve. Reading is the preferred path rather than the only one: replacing the value’s characters with unreadable dots, leaving attention untouched, also removes following; the Qwen models then reconstruct the value from the previous line in 0.60 to 0.73 of cases, Phi-3-medium never (0.00).

5.2 Where retrieval occurs across layers

Layer-resolved versions of both interventions locate the read in depth on Qwen3-4B (36 layers; Figure 4B, C). Overwriting the value’s hidden state at any single layer up to 22 redirects the answer at 0.90 to 0.94; at layer 24 and beyond, at 0.02 or less ($n = 60$): the value can be rewritten at any layer before the read and at none after. Blocking the value-position attention only at the middle third of layers (12 through 23) collapses following from 0.857 to 0.025, blocking only the early or late third leaves it unchanged, and no single layer’s block matters ($n = 80$): the read is distributed redundantly across the middle third and confined to it; the two sweeps agree on the same boundary, near layer 23. Coarse sweeps on two further models (appendix) show the same structure with model-specific depth.

6 When models retrieve versus recompute

By default, models reuse the value they wrote rather than recompute it. On DeepSeek V3.2, an edited value the prompt merely implies is followed at 0.93, while an edited value whose derivation the prompt states is reverted at 0.43; with the defining sentence present but the edit two steps away, following returns to 0.96, so the reversion targets the defined value. Claude Sonnet 4.5 recomputes more broadly, reverting most when a stated checkpoint is the value’s only source. Llama 3.3 70B follows edits at 0.97 either way, despite solving the same chains from scratch: the capacity to recompute does not imply recomputing. Qwen models at 4B and 7B never revert. The conditions are separate experiments; we claim within-experiment contrasts only.

On GSM8K worked solutions, an edited intermediate is followed at 0.87 by Qwen3-4B and at 0.10 by V3.2: a benchmark intermediate is often one operation from numbers still in the prompt. Within one model, V3.2 follows edits at 0.93 on chains it cannot regenerate and at 0.10 on GSM8K. Lanham et al. [2023] report reliance on written reasoning declining with scale on standard benchmarks; our contrasts indicate the direction depends on the state. Faithfulness is a property of the state’s regenerability: on regenerable state, stronger models depend less on the trace; on state no model can regenerate, dependence is high at every scale.

7 Related work

Our interventions build on causal tracing and activation patching [Meng et al., 2022] and causal abstraction [Geiger et al., 2021]; Heimersheim and Nanda [2024] caution that broad patches conflate a component mattering with deleting information at a position, which the single-layer sweep in Section 5 addresses. Faithfulness studies [Turpin et al., 2023, Lanham et al., 2023] and filler-token results [Pfau et al., 2024] bound our claims to state the model does not regenerate internally.

8 Limitations

Tasks are numeric and count state tracking, so our claims concern serial task state; code, plans, and multi-entity state are untested. The J-space result bounds one fitted lens at generation-preserving doses and does not measure the J-space content of the written positions. The fine layer sweeps cover one model; the coarse cross-model sweeps do not resolve Qwen2.5-7B’s read boundary. White-box evidence is at 4B to 14B scale, frontier evidence behavioral. Conditioning on edit-following (Section 4) makes cross-model levels not directly comparable; the appendix reports unconditioned rates. The only patch control is size-matched noise; a structured control planting a valid activation for a different quantity would further isolate value content.

9 Conclusion

Serial reasoning state outgrows what a language model’s internal workspace supports within one to two dependent steps, and no intervention shifts state between workspace and trace: the chain of thought is working memory rather than overflow, functioning as register-like memory, settable by intervention, retrieved through middle-layer attention, independently addressable, and decisive for the answer. For oversight, identical text can be made to carry different stored values, so monitoring sees the channel, not the state; whether they arise naturally is untested.

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